

MAT 3150 — Homework 4

Due October 1

Problem 1 (10 points)

Let $(x_n)_n$ and $(y_n)_n$ be two sequences of real numbers, so that $(y_n)_n$ is increasing. Assume that

$$\lim_{n \rightarrow \infty} \frac{x_{n+1} - x_n}{y_{n+1} - y_n} = L,$$

for some $L \in \mathbb{R}$. Prove that for any $\varepsilon > 0$ there exists $N > 0$ so that

$$(L - \varepsilon)(y_{n+1} - y_N) < x_{n+1} - x_N < (L + \varepsilon)(y_{n+1} - y_N),$$

for all $n > N$.

Hint: Use $x_{n+1} - x_N = (x_{n+1} - x_n) + (x_n - x_{n-1}) + \cdots + (x_{N+1} - x_N)$, for any $n > N$.

Proof. Let $\varepsilon > 0$.

Step 1: Use the hypothesis. [TODO: What does $\lim_{k \rightarrow \infty} \frac{x_{k+1} - x_k}{y_{k+1} - y_k} = L$ give for this ε ?]

Step 2: Clear the denominator. [TODO: Rewrite the inequality from Step 1 without a fraction. Justify why the inequality direction is preserved.]

Step 3: Sum from $k = N$ to $k = n$. [TODO: Apply Step 2 to each k and combine, using the hint.]

Step 4: Simplify. [TODO: What do the left and right sides become?]

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Problem 2 (10 points)

Let $(x_n)_n$ and $(y_n)_n$ be two sequences of real numbers, so that $(y_n)_n$ is increasing and $\lim_{n \rightarrow \infty} y_n = \infty$. Assume that

$$\lim_{n \rightarrow \infty} \frac{x_{n+1} - x_n}{y_{n+1} - y_n} = L,$$

for some $L \in \mathbb{R}$. Prove that

$$\lim_{n \rightarrow \infty} \frac{x_n}{y_n}$$

exists and is also equal to L .

Hint: Use Problem 1.

Proof. Let $\varepsilon > 0$, and let N be as in Problem 1.

Step 1: Rearrange Problem 1's inequality. [TODO: Isolate x_{n+1} and divide so that $\frac{x_{n+1}}{y_{n+1}}$ is in the middle. Why is $y_{n+1} > 0$ for large n ?]

Step 2: Handle the terms involving N . [TODO: Which terms are “leftovers” with x_N, y_N ? Why do they tend to 0?]

Step 3: Choose the final threshold. [TODO: Pick M so that both estimates hold, and conclude $\left| \frac{x_n}{y_n} - L \right| < \underline{\hspace{1cm}}$ for $n \geq M$.]

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Problem 3 (10 points)

Prove that if

$$\lim_{n \rightarrow \infty} a_n = L,$$

then

$$\lim_{n \rightarrow \infty} \frac{a_1 + a_2 + \cdots + a_n}{n}$$

exists and equals L .

Proof. [TODO: Choose $x_n = \underline{\hspace{1cm}}$ and $y_n = \underline{\hspace{1cm}}$.]

[TODO: Check the hypotheses of Problem 2: (y_n) increasing, $y_n \rightarrow \infty$.]

[TODO: Compute $\frac{x_{n+1} - x_n}{y_{n+1} - y_n}$ and find its limit.]

[TODO: Conclude.]

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Problem 4 (10 points)

Define the sequence $a_1 = 1$ and

$$a_{n+1} = 3 - \frac{1}{a_n}.$$

Show that a_n is convergent, by proving that a_n is monotone and bounded. What is the limit of a_n ?

Claim. (a_n) is $\underline{\hspace{1cm}}$ (increasing / decreasing) and $\underline{\hspace{1cm}} \leq a_n \leq \underline{\hspace{1cm}}$ for all n .

Proof. **Step 1: Bounded.** [TODO: Base case.] [TODO: Inductive step: assume $\underline{\hspace{1cm}} \leq a_n \leq \underline{\hspace{1cm}}$, show the same for a_{n+1} .]

Step 2: Monotone. [TODO: Base case: compare a_1 and a_2 .] [TODO: Inductive step.]

Step 3: Convergence. [TODO: Cite the Monotone Convergence Theorem.]

Step 4: The limit. [TODO: Let $a = \lim a_n$. Why is $a \neq 0$? Pass to the limit in $a_{n+1} = 3 - \frac{1}{a_n}$ and solve.] [TODO: Which root, and why?]

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